ABDULLAH AL KHALID

Software Engineer / Data & Distributed Systems
Dhaka, Bangladesh
khalidrafi1111@gmail.com ◆ +8801847791909
linkedin.com/in/khalidrafi1111 ◆ github.com/An1MuS1111

SUMMARY

Software engineering graduate focused on distributed systems, databases, and backend infrastructure. Experience with .NET APIs, SQL data pipelines, and systems work in Rust and Go — including Raft, gossip protocols, CRDTs, and append-only logs.

WORK EXPERIENCE

DKSH CSSC **Kuala Lumpur, Malaysia**
Software Engineer Intern Sep 2024 – Feb 2025
Worked with cross functional teams to keep production applications running smoothly, building a website plugin, automating manual work, and migrating SQL Server data through .NET Core APIs and pipelines.

EDUCATION

Universiti Teknologi Malaysia (UTM) **Skudai, Johor, Malaysia**
Bachelor of Computer Science with Honours in Software Engineering Mar 2021 – Mar 2026

PASSION PROJECTS

ironbarrel (Rust)
A log-structured hash table key-value storage engine inspired by the orignal bitcask

- Built a high throughput storage engine from ground up in Rust, inspired by the Basho Bitcask paper; designed lock-free concurrent indexing (KeyDir) and positional reads (pread) to achieve 280k+ ops/sec with 1.2µs p50 read latency.
- Engineered a crash-resilient storage & compaction pipeline using hardware-accelerated CRC32 and binary hint-file indexing, cutting 1M-key startup recovery by 96% (from 2.4s to 80ms) while automatically reclaiming dead disk space.

Miserable Messengers / Gossip Gloomers (Rust, Maelstrom/Jepsen)
Fly.io distributed systems challenge series

- Built a Rust node engine that solved Fly.io distributed systems challenges, including gossip broadcast, CRDT counters, and a linearizable KV store.
- Verified correctness under network partitions, packet drops, and concurrent clients using Jepsen's Maelstrom;

ACTIVITIES & COMPETITIONS

RecyclO · Top 5 **Honeywell STEM Challenge**
Plastic Waste Management System 2023
Designed a plastic waste management system aimed at reducing plastic waste through practical collection and processing workflows. Focused on collaborative, sustainable solutions intended to reduce environmental impact.

Flood Remission System · Top 5 **App Innovation Challenge**
Disaster Management 2022
Built a flood-mitigation app that forecasts flooding from live weather-sensor and satellite data, sends early warnings to residents, and helps authorities coordinate rescue and shelter.

REFERENCES

A. S. M. Khairuzzaman
Managing Director & CEO (Acting); Deputy Managing Director & COO
Central Counterparty Bangladesh PLC
DSE Tower, Level 5, Plot-46, Road-21, Nikunja-2, Dhaka-1229, Bangladesh
+8801713425801 · dmd@ccbl.com.bd

Abdullah Al Noman
Solution Architect
DKSH CSSC
60000 Kuala Lumpur, Wilayah Persekutuan Kuala Lumpur, Malaysia
+60 11-2811 8959 · md.abdullah.noman@dksh.com

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